

short. In such cases, extending Drupal is easy and all one has to do is to roll out one's own module.

Security

Drupal is as secure and sometimes more secure than any other competitive commercial offering. It is built on a set of APIs that mitigate common security risks. Drupal addresses the OWASP (Open Web Application Security Project-https://www.owasp.org/index.php/Main_Page) top 10 security risks:

- SQL injection hacks
- Cross-site scripting
- Broken authentication and session management
- Insecure direct object reference
- Cross-site request forgery
- Security misconfiguration
- Failure to restrict URL access
- Non-validated redirects and forwards
- Insecure cryptographic storage
- Insufficient transport layer protection

These are very compelling reasons to ensure that your next site is built on Drupal. It is for these very reasons that the White House (<http://www.whitehouse.gov/>), *The Economist* magazine (<http://www.economist.com/>) and several others have chosen Drupal.

(To determine the stack that a site runs on you can visit the site and view the technology profile using the 'built with' add-on for Firefox.)

Drupal's flexible architecture

Drupal provides Web developers several pre-built components to get started and a set of tools to build a comprehensive content management solution. In short, Drupal provides a set of building blocks to a Web developer to rapidly assemble a site solution.

In Drupal, 'nodes' hold all the information pertaining to the content like the title, body, author, published date, etc. The menu system and the taxonomy provide the information architecture. The presentation and display is handled by the theme layer.

This makes it possible for the Web development team to simultaneously make changes to several functions of the site without affecting the other development teams' ability to propagate their changes.

Ease of administration

Every visitor who lands on your site is considered a 'user' by Drupal. To differentiate between different types of users and determine their entitlements on the site, all those who have a user ID in Drupal are non-anonymous users and those who don't are called anonymous users. You can create various classes of users by defining roles and assigning users to them. You can then assign permissions to these roles so that users who are assigned those roles inherit those permissions.

Adding content to a site is as easy as sending out an email. Provide a title and body of the content and click 'Save' and your

content is published. When you create content, you can choose to include an image, ask Drupal to automatically create a menu link, provide a URL path where the page is accessible, decide to accept comments from visitors, etc. You can also decide to publish the page or save it as a draft to revisit it at a later date.

However, don't let this ease mislead you into thinking that Drupal doesn't provide enough for the power user. You can build your site's functionality by enabling contributed modules. There are modules available for almost every aspect of a site's functionality, like:

- Actions, workflows and triggers
- CRM systems
- Advertising and commerce
- Events
- Search engine optimisation

...and many more. If you at any time feel that these modules fall short of providing you the features you require, you can roll out your own module.

Not really impressed? Well, then you can create a new Drupal distribution if you wish. A Drupal distribution includes the Drupal core along with the themes, modules, libraries and an installation profile. At the time of writing this, there were no other CMSs that allowed you to do this.

Developing for mobile devices

Drupal provides a layer of abstraction between native mobile apps (residing on the device) and the Drupal CMS. Mobile apps can talk to Drupal using the 'services' module to fetch content, validate users, execute business logic, etc. If you instead wish to develop a Web app that resides on your Web server, you can integrate it with the Drupal theme layers. You can use any of the available libraries like JQTouch, Enyo JS, Sencha Touch, jQTouch, Jo, jqMobi, etc.

If you are planning on building a site for a specific device, there are modules to detect your visitors' device and redirect them to the respective URLs.

Several organisations across industry segments have built their sites on Drupal. Case studies can be found on the Drupal.org website at <https://drupal.org/case-studies>. These sites cover everything from the arts to travel and hospitality. These case studies describe the challenges and provide solutions on how to tackle the complexities of building a complex website.

Considering that Drupal provides several of the building blocks required to assemble a site, the depth of the knowledge available and the large user community, it is worthwhile considering the Drupal platform for your next corporate site or Web solution. 

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